

Probability and Statistics – 1st Semester

Total: 20 questions (20 points).

Duration: 1 HOUR

Use of phones, documents, or any form of cheating is strictly prohibited. Any student caught cheating will receive a grade of 0.

Instructions: Choose **ONE** correct answer for each question.

- 1 pt for each good answer
- -0.25 pt for a wrong answer
- 0 pt for no answer

Q1. A continuous random variable:

- A. Takes only integer values
- B. Has $P(X=x)=0$ for any single value
- C. Has a probability mass function
- D. Is always uniform

Q2. The covariance between two independent variables is:

- A. Always 1
- B. Always 0
- C. Always negative
- D. Undefined

Q3. The correlation coefficient:

- A. Can be greater than 1
- B. Is always positive
- C. Lies between -1 and 1
- D. Equals variance

Q4. The Central Limit Theorem states:

- A. The sample mean becomes uniform
- B. The sample mean becomes normal for large n
- C. Variance goes to zero
- D. Data becomes discrete

Q5. Let X have density $f(x)=kx$ on $[0,2]$. The value of k is:

- A. $1/2$
- B. $1/4$
- C. 1
- D. 2

Q6. $E[X]$ is:

- A. 1
- B. $4/3$
- C. 2
- D. $3/2$

Q7. $P(X \leq 1)$ is:

- A. $1/4$
- B. $1/2$
- C. $3/4$
- D. $2/3$

Q8. A valid density must:

- A. Be increasing
- B. Integrate to 1
- C. Be symmetric
- D. Be continuous

Q9. If $F(x)$ is a CDF:

- A. It decreases
- B. It is bounded between 0 and 1
- C. It can exceed 1
- D. It is always linear

Q10. $P(X=0, Y=0)=0.2$, $P(0,1)=0.3$, $P(1,0)=0.1$, $P(1,1)=0.4$. $P(X=1)$ is:

- A. 0.3
- B. 0.4
- C. 0.5
- D. 0.6

Q11. X and Y are:

- A. Independent
- B. Not independent
- C. Always correlated
- D. Always independent

Q12. $E[X]$ is:

- A. 0.4
- B. 1
- C. 0.6
- D. 0.5

Q13. If $X \sim N(100, 10^2)$, then $Z=(X-100)/10$ follows:

- A. Uniform
- B. Normal(0,1)
- C. Exponential
- D. Binomial

Q14. $P(X > 110)$ is approximately:

- A. 0.5
- B. 0.1587
- C. 0.8413
- D. 0.025

Q15. For $n=25$, standard deviation of $\bar{X}$ is:

- A. 10
- B. 5
- C. 2
- D. 1

Q16. Distribution of $\bar{X}$:

- A. Uniform
- B. Normal
- C. Bernoulli
- D. Poisson

Q17. The likelihood function:

- A. Is always constant
- B. Depends on parameter θ
- C. Is a probability of parameters
- D. Is always linear

Q18. MLE is:

- A. Minimizes likelihood
- B. Maximizes likelihood
- C. Equals mean always
- D. Equals variance

Q19. A fair die is rolled. $E[X]$ is:

- A. 3
- B. 3.5
- C. 4
- D. 2.5

Q20. $P(X \geq 5)$ is:

- A. $1/6$
- B. $1/3$
- C. $1/2$
- D. $2/3$